

Detector Data

Background Data
(Training)

Calibration Data
(Training)

Preprocess Data to
Extract Input Features

Train SOM

Cluster
Boundary
Establishment

Interpret Clusters and
Label into Physical
Classes

Calibration Data

Simulation Data

Synthetic
Data

Event Validation

NR Band
Validation

ER Band
Validation

Argon
Validation

Simulation
Validation

Classification
Accuracy

Efficiency Bias
Study

AC Validation

Compare Classification to Previous Work